FINANCE PORTAL · DOCUMENTATION

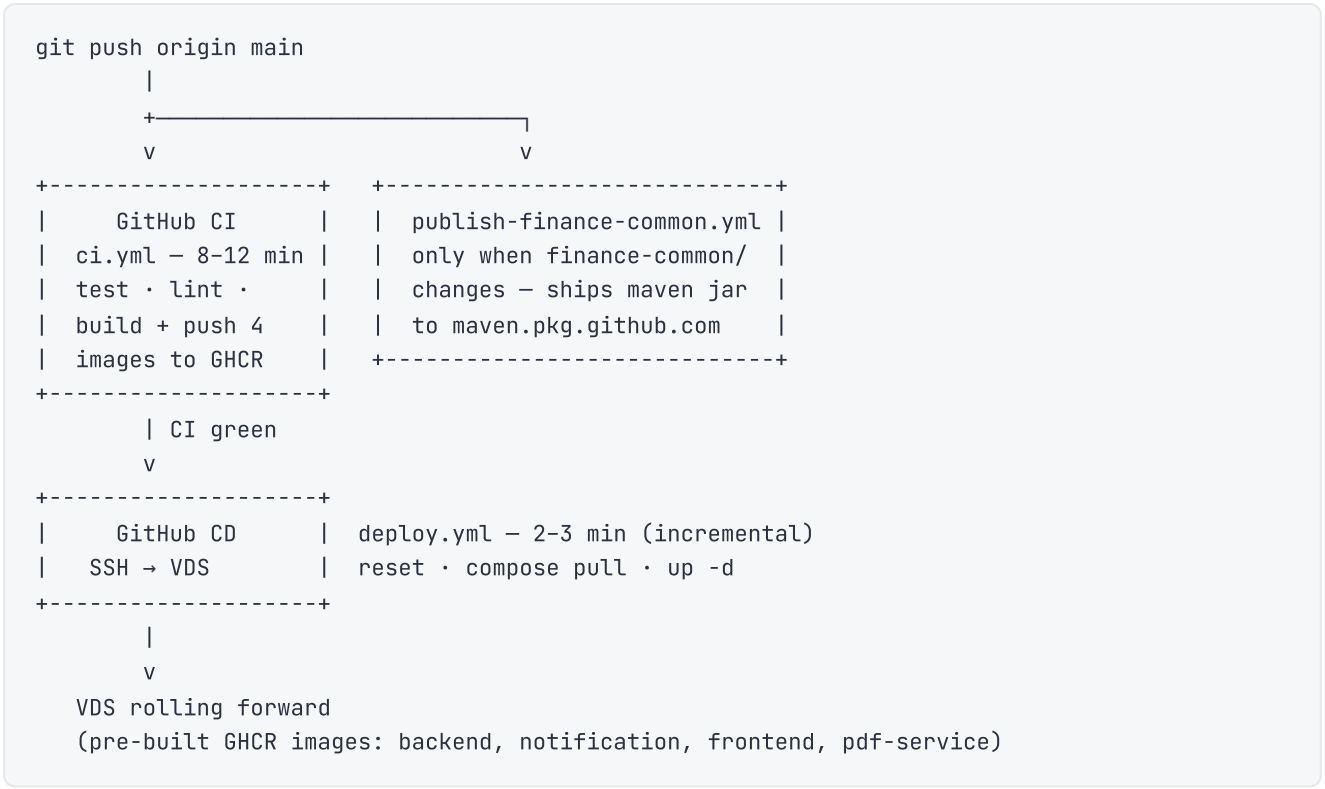
Deployment & CI / CD

How CI tests + builds images on every push, how CD auto-deploys from a green main, the required secrets, and the manual / rollback operations.

End-to-end guide: from a blank VDS to a live, self-deploying production stack. Once it's set up, every `git push origin main` ends with the new version rolling on the server about fifteen minutes later — no manual SSH.

What this covers

End-to-end flow



End-to-end: ~15 minutes from `git push` to the new version live for an incremental update. After the initial setup, no manual SSH is ever needed.

Prerequisites — accounts to create

SERVICE	WHAT FOR	RECOMMENDED	COST
Domain + DNS + TLS	one place for everything	**Cloudflare** (registrar + DNS + free proxy TLS)	~$10/year
VDS (compute)	running the stack	**Contabo Cloud VPS** (16 GB+ x86, EU) or Hetzner	~$9–25/month
Mail relay (SMTP)	outbound mail (verify, 2FA, alerts)	**Brevo** (300/day free ≈ 9k/month) or Resend	$0 for low volume
GitHub	code + CI + image registry	GitHub (GHCR comes with the repo)	$0

Total typical cost: **~$10/year + $9–25/month**.

First-time VDS setup

A one-time, ~1-hour walkthrough done from scratch on a fresh server. After this the CD pipeline handles everything.

1. Provision the VDS

- **Spec**: ≥16 GB RAM (the stack runs OpenSearch + Kafka + Keycloak + two JVMs + Postgres), x86/amd64, Ubuntu 24.04 LTS. ~100 GB NVMe is plenty for code + Postgres + indices + image cache. EU region for low latency from TR.
- Order from your chosen provider (Contabo Cloud VPS 20 NVMe and similar tiers fit the bill).
- During order, paste your laptop's SSH public key (next step) into the provider's "SSH public key" field — most providers offer this and skip generating a root password.

2. SSH access from your laptop

Generate a key once on your machine (no passphrase — required for unattended deploys):

```
ssh-keygen -t ed25519 -C "your-comment" -f ~/.ssh/id_ed25519 -N ""
cat ~/.ssh/id_ed25519.pub          # paste this into the provider
```

After the VDS is provisioned, verify the key login works:

```
ssh root@<VDS_IP> 'uname -a'
```

If the provider auto-generated a password instead of accepting the key, install it once with that password:

```
ssh-copy-id -i ~/.ssh/id_ed25519.pub root@<VDS_IP>
```

3. Initial hardening — disable password authentication

Once key login works, drop the password door entirely. Bots constantly brute-force SSH on the open internet — turning passwords off closes that attack surface.

On the server, in `/etc/ssh/sshd_config`, set `PasswordAuthentication no`. Ubuntu cloud images usually re-enable it inside `/etc/ssh/sshd_config.d/50-cloud-init.conf` — change it there too. Then `systemctl reload ssh`. Verify with `sshd -T | grep -i passwordauth`.

Your key login keeps working; anything else is rejected. **Don't lose the private key on your laptop** — back it up to a password manager or Time Machine.

4. Install Docker + Compose plugin

The official one-shot script (Ubuntu / Debian / RHEL):

```
ssh root@<VDS_IP> 'curl -fsSL https://get.docker.com | sh'
ssh root@<VDS_IP> 'docker --version && docker compose version'
```

5. Clone the repo on the VDS

The repo is public, so a plain HTTPS clone works — no deploy key or token needed:

```
ssh root@<VDS_IP>
cd /root
git clone https://github.com/<owner>/<repo>.git
cd <repo>
```

> If you later make the repo private, swap the HTTPS URL for SSH (`git@github.com:<owner>/<repo>.git`) and add a read-only **Deploy Key** (Settings → Deploy keys) so the VDS can pull without a user account.

6. Cloudflare DNS + free HTTPS

Cloudflare gives DNS + TLS termination for free if your domain is on its DNS.

In the Cloudflare dashboard → `<your-domain>` → **DNS** → **Records** → **Add record**:

TYPE	NAME	VALUE	PROXY
A	`@`	`<VDS_IP>`	**Proxied** (orange cloud)
A	`www`	`<VDS_IP>`	**Proxied**

The orange-cloud proxy gives you free HTTPS at the edge (no Let's Encrypt to manage), DDoS protection, and a CDN. Your VDS keeps serving plain HTTP on port 80; Cloudflare terminates TLS.

7. Mail relay — Brevo example

Brevo's free tier (300/day ≈ 9,000/month) is plenty for transactional mail (account verify, 2FA, alerts). Same flow for Resend / SES with minor differences.

1. Sign up at `app.brevo.com` (no card needed for free).
2. **Senders & Domains** → **Add domain** → enter `your-domain.com`.
3. Brevo gives you 2-3 DNS records to verify (DKIM CNAMEs). If your DNS is on Cloudflare, Brevo can install them automatically with one click ("Authorize") — much easier than pasting them by hand.
4. Wait for verification to flip green (1-10 minutes).
5. **Settings** → **SMTP & API** → **Generate a new SMTP key**. Copy the long key once — Brevo won't show it again.

You'll plug `smtp-relay.brevo.com:587` + your Brevo SMTP login + that key into the `.env` in the next step.

8. Production `.env`

The repo ships a `.env.example` with every variable documented. Copy and fill in:

```
cd /root/<repo>
cp .env.example .env
chmod 600 .env
nano .env
```

Replace the placeholders with strong random values from your password manager (or `pwgen`). **OpenSearch needs 8+ characters with upper, lower, digit, and a special character** — respect that or the container won't start.

Critical sections:

- **Login**: `FINANCE_ADMIN_PASSWORD`, `DEMO_USER_PASSWORD` — the demo-visible accounts (keep simple if it's a showcase; randomise for a real install).
- **Infra**: `KEYCLOAK_ADMIN_PASSWORD`, `DB_PASSWORD`, `REDIS_PASSWORD`, `LDAP_ADMIN_PASSWORD`, `LDAP_CONFIG_PASSWORD`, `OPENSEARCH_PASSWORD` (OpenSearch needs `≥ 8` chars with upper + lower + digit + special — the inline command above produces a compliant one).
- **Mail**: `SPRING_MAIL_HOST=smtp-relay.brevo.com`, `SPRING_MAIL_PORT=587`, `SPRING_MAIL_USERNAME=<brevo SMTP login>`, `SPRING_MAIL_PASSWORD=<brevo SMTP key>`, `SPRING_MAIL_PROPERTIES_MAIL_SMTP_AUTH=true`, `SPRING_MAIL_PROPERTIES_MAIL_SMTP_STARTTLS_ENABLE=true`, `NOTIFICATION_FROM_ADDRESS=noreply@your-domain.com`
- **Domain / frontend**: `FRONTEND_BASE_URL=https://your-domain.com`, `VITE_KEYCLOAK_URL=https://your-domain.com/auth`. These are baked into the frontend at build time — set them before the first build.

Demo / showcase tip: with `make demo` or the `docker-compose.demo.yml` overlay you get a seeded portfolio and a `demouser` account; in that mode you don't need any market-data API keys (`EVDS_API_KEY` / `COINGECKO_KEY` can be empty). Set `APP_MARKET_INIT_ENABLED=false` so the backend doesn't try a cold-start fetch.

9. First deploy

The first deploy builds the images on the server (CI hasn't pushed any yet). Use the demo overlay if you want seeded data; layer the prod overlay on top to skip the Mailpit dev catcher and send mail through Brevo:

```
docker compose -f docker-compose.yml -f docker-compose.demo.yml -f docker-compose.prod.yml up -d --build
```

First build is ~25–35 minutes (Maven downloads + JVM compile + Vite + Docker layers + db-seed loading the 46 MB market snapshot). Watch it with `docker compose logs -f`.

10. Verify it's up

```
docker compose ps                                       # everything Up + healthy
curl -fsS https://your-domain.com/actuator/health       # should be 200
curl -fsS https://your-domain.com/                      # frontend HTML
```

Open `https://your-domain.com` in a browser. Login with the admin credentials from `.env`. Trigger a real action (price alert / 2FA / account verify) and watch the email show up at your real inbox (it goes out through Brevo with proper DKIM, not Mailpit).

That's the one-time setup done. From here on, every `git push origin main` deploys automatically.

CI pipeline (`.github/workflows/ci.yml`)

Triggered on every pull request and every push to `main`.

JOB	WHAT IT DOES	DEPENDS ON
`finance-common-test`	builds + tests the shared library, uploads JaCoCo XML	—
`backend-test`	installs finance-common, runs the backend Maven reactor (`./mvnw verify`), uploads per-module JaCoCo XML	finance-common-test
`notification-test`	builds + tests the notification microservice, uploads JaCoCo XML	finance-common-test
`frontend-lint`	`npm ci && npm run lint`	—
`sonar`	SonarQube scan, downloads JaCoCo artifacts (gated on `vars.SONAR_HOST_URL`)	all the above
`docker-backend`	builds + pushes the backend image to GHCR	backend-test, finance-common-test
`docker-notification-backend`	same for notification	notification-test, finance-common-test
`docker-frontend`	same for frontend (with `VITE_*` build args from repo variables)	frontend-lint
`docker-pdf-service`	builds + pushes the PDF render service image (Node / Puppeteer)	—
`compose-validate`	sanity-checks the merged compose resolves cleanly	all docker-*

Tests + lint run on **every PR**. Image builds and the push to GHCR only run on **push to** `main`, so PRs never pollute the registry. JaCoCo XML is uploaded as a workflow artifact so the Sonar job can read

it across modules. Buildx layer cache (`type=gha`) is reused per image so subsequent builds typically finish in 1–3 minutes per image instead of 8–12.

Publish finance-common workflow (`.github/workflows/publish-finance-common.yml`)

`finance-common` is a Maven library shared between the two backends. CI builds the JAR from source inside each backend's Docker image, so day-to-day development never needs the published artifact. This separate workflow exists to keep a versioned copy on **GitHub Packages (Maven)** for external consumers (or other repos in the same org).

```
on:
  push:
    branches: [main]
    paths:
      - 'finance-common/**'
      - '.github/workflows/publish-finance-common.yml'
  workflow_dispatch:
```

The `paths:` filter means the workflow only runs when `finance-common/` actually changes (or when manually dispatched) — most pushes skip it entirely. The job runs `mvn -B deploy` against `maven.pkg.github.com` using the workflow's auto-issued `GITHUB_TOKEN`, so no PAT is needed. The published artifact appears under the repo's **Packages** → **Maven** section.

CD pipeline (`.github/workflows/deploy.yml`)

Triggered automatically when CI succeeds on `main`, or manually via the Actions UI:

```
on:
  workflow_dispatch:
  workflow_run:
    workflows: ["CI"]
    types: [completed]
    branches: [main]

jobs:
  deploy:
    if: github.event_name == 'workflow_dispatch'
        || github.event.workflow_run.conclusion == 'success'
```

The single job uses `appleboy/ssh-action@v1` to open an SSH session to the VDS with the key from the `VDS_SSH_KEY` secret. Once connected it runs:

1. `git fetch origin main && git reset --hard origin/main && git clean -fd` — replaces the working tree with the latest `main`, dropping any local edits or rsync leftovers. This is defensive: a previous manual change on the server can no longer block the pull.
2. `docker compose pull` — fetches every image referenced by the merged compose (the four GHCR ones plus the public third-party images: Postgres, Kafka, Keycloak, OpenSearch, Redis,

Nginx, ...).

3. `docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d` — rolling restart of only the services whose image hash actually changed. Containers with the same hash keep running, so deploys are near-zero-downtime for the unchanged services (gateway, DB, etc.) and brief for the ones that did change.
4. `docker image prune -f` — removes the now-orphaned previous image versions so disk doesn't grow forever.

No source compile on the server, no Maven, no npm — only Docker. Pull + restart only. ~2–3 minutes for an incremental update, longer on the first deploy when the image cache is empty.

Mail — choosing the FROM address

The mail provider (Brevo / Resend / SES) is the SMTP relay; it isn't a mailbox. You don't need to "create an account" for `noreply@your-domain.com` — once the domain is verified in the provider, you can send from **any address on that domain** and the provider signs it with DKIM so receivers accept it as legitimate.

Common conventions:

ADDRESS	USE
`noreply@your-domain.com`	Signals "don't reply, this is automated". Most common.
`hi@your-domain.com`	Friendlier, e.g. for product newsletters.
`admin@your-domain.com`	System / ops alerts.

Set the chosen address in `NOTIFICATION_FROM_ADDRESS` in `.env`. To **receive** replies you need actual email hosting (Google Workspace, Zoho, Fastmail ...) — that's a separate concern from sending and isn't required for the app to work.

Image registry

CI builds three custom images and pushes them to **GitHub Container Registry (GHCR)**, tagged with both `latest` and the short commit SHA:

```
ghcr.io/<owner>/finance-portal-backend
ghcr.io/<owner>/finance-portal-notification-backend
ghcr.io/<owner>/finance-portal-frontend
```

The packages are public on GHCR (the repo is public), so the VDS pulls without any registry login. All third-party images (Postgres, Kafka, Keycloak, OpenSearch, Redis, Nginx, ...) come from their public official registries unchanged.

Required GitHub repo secrets

Under **Settings → Secrets and variables → Actions**:

SECRET	PURPOSE
`VDS_HOST`	Server IP or hostname
`VDS_USER`	SSH user on the server
`VDS_SSH_KEY`	Private key matching the server's `~/.ssh/authorized_keys`
`VDS_APP_DIR`	Absolute path of the cloned repo on the server (e.g. `/root/<repo>`)

The GHCR images are public, so the deploy workflow pulls them without any registry token. No PAT is needed.

Optional, only with self-hosted Sonar:

- `SONAR_TOKEN` (secret) and `SONAR_HOST_URL` (variable)

Manual operations

Force a deploy from any commit without touching code:

```
gh workflow run deploy.yml
```

Or: Actions → *Deploy to VDS* → **Run workflow**.

Pin a specific image version (rollback to a known-good SHA): on the server, set `BACKEND_IMAGE=ghcr.io/.../finance-portal-backend:<sha>` in `.env` and run `docker compose -f docker-compose.yml -f docker-compose.prod.yml up -d backend`. The compose file reads `${BACKEND_IMAGE:-…}`, so any tag can be pinned through env without editing source.

Check what's running:

```
ssh root@vds 'cd app && docker compose ps'
ssh root@vds 'cd app && docker compose logs -f --tail=200 backend'
```

Stop the stack temporarily (to save VDS cost):

```
ssh root@vds 'cd app && docker compose down'   # keeps data
```

Failure handling

- A failing test, lint, or Sonar quality gate **fails CI** and stops the pipeline — no image is pushed, no deploy fires. The running version on the VDS is untouched.

- A failing image build (`docker-*`) likewise blocks deploy. CI must be green for `deploy.yml` to fire.
- If `deploy.yml` itself fails (SSH timeout, registry login error, compose error), the previous version keeps running on the VDS — `docker compose pull` only pulls; the running container is replaced only when `up -d` succeeds.
- A container failing to start after `up -d` is visible in `docker compose ps` and the container's logs (`docker compose logs <service>`). Roll back to the previous SHA with the image-tag override above.

Why this shape

- CI tests + image build are split into separate jobs so the slow ones (Maven reactor, Docker buildx) run in parallel with the fast ones (lint, finance-common). Failures surface faster.
- CD is a separate workflow (`workflow_run`) instead of one more job in CI, so deploys honour CI's exact pass/fail and never race the image push.
- Images are pre-built in CI rather than on the server: the VDS doesn't need Maven, npm, build tools or large heap — only Docker. Deploys are fast, the server stays small, and every environment runs the exact same image bytes that CI tested.
- Cloudflare in front of the VDS gives free HTTPS, free DDoS protection, and a CDN cache for the static frontend — without any TLS certificate to manage on the server.
- One single SMTP block in `.env` drives **both** the app's transactional mail and Keycloak's account emails (verify / 2FA / reset), so switching mail provider is one config block to update, not two.

Local vs VDS — what's actually different

Same compose, same code, same images. Only a few `.env` keys change:

KEY	LOCAL DEFAULT	VDS VALUE
`FRONTEND_BASE_URL`	`http://localhost`	`https://your-domain.com`
`VITE_API_BASE_URL`	`/api/v1` (same-origin)	`/api/v1` (same-origin, unchanged)
`VITE_KEYCLOAK_URL`	`http://localhost/auth`	`https://your-domain.com/auth`
`SPRING_MAIL_HOST`	`mailpit` (catcher)	`smtp-relay.brevo.com`
`SPRING_MAIL_PORT`	`1025`	`587`
`SPRING_MAIL_USERNAME`	(blank)	your Brevo SMTP login
`SPRING_MAIL_PASSWORD`	(blank)	your Brevo SMTP key

KEY	LOCAL DEFAULT	VDS VALUE
`SPRING_MAIL_PROPERTIES_MAIL_SMTP_AUTH`	`false`	`true`
`SPRING_MAIL_PROPERTIES_MAIL_SMTP_STARTTLS_ENABLE`	`false`	`true`
`NOTIFICATION_FROM_ADDRESS`	`noreply@finance.local`	`noreply@your-domain.com`
`ORIGIN_CERT_CN`	unset (defaults `localhost`)	`your-domain.com`
`APP_MARKET_INIT_ENABLED`	`true`	`false` (demo seed is loaded instead)

The compose overlay also changes: `docker-compose.yml` + `docker-compose.prod.yml` on VDS (drops the Mailpit dev catcher); `docker-compose.yml` + `docker-compose.demo.yml` locally (adds the demo seed and `demouser`).

Choosing demo vs empty database

Two ways to bring the stack up — same code, different startup seed:

- **Demo (`make demo`)** — `docker-compose.demo.yml` adds the `db-seed` one-shot container that loads `seed/market_data.sql.gz` (~4 years of candles) plus `seed/demo_portfolio.sql` (the sample portfolio), and the `realm-demo.json` realm provisions `demouser` / `Demo123!` who owns that portfolio. Best for clone-and-run showcase. No EVDS/CoinGecko key needed.
- **Empty (`make empty`)** — boots with empty tables. The backend's `MarketDataInitializer` detects empty tables and pulls multi-year history from the upstreams (FX → stocks → commodities) on first run; this takes ~20–30 min and **requires** `EVDS_API_KEY` because stocks and commodities are converted through FX. Set `APP_MARKET_INIT_ENABLED=false` to skip the cold-start fetch entirely.

`make reset` wipes all volumes if you want to switch modes; `make down` just stops without losing data.